AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT132 – M132
Linear Algebra

Final Examination
Fall Semester 2024/2025

Date: 09 / Jan. /2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

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Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Q–1: [5×3 marks] Choose the correct answer:

1.1: The linear system $\begin{cases} x+2y=4 \\ 4x+8y=-16 \end{cases}$

a) has no solution
b) has an infinitely many solutions
c) has a unique solution $X=\begin{bmatrix}1\\2\end{bmatrix}$
d) has a unique solution $X=\begin{bmatrix}2\\1\end{bmatrix}$
e) None of the above

1.2: The matrix $A=\begin{bmatrix}a & 4\\ 2 & a-2\end{bmatrix}$ is singular at

a) $a=2$
b) $a=-4$
c) $a=2,-4$
d) $a=-2$
e) None of the above

1.3: The dimension of the subspace $W=\left\{\begin{bmatrix}a\\2a\\3a\end{bmatrix}: a\in\mathbb{R}\right\}$ is

a) 0
b) 1
c) 2
d) 3
e) None of the above

1.4: A spanning set for $W=\left\{\begin{bmatrix}2a\\a\end{bmatrix}; a\in R\right\}$ is

a) $\left\{\begin{bmatrix}2\\0\end{bmatrix},\begin{bmatrix}0\\1\end{bmatrix}\right\}$
b) $\left\{\begin{bmatrix}2\\1\end{bmatrix}\right\}$
c) $\left\{\begin{bmatrix}1\\2\end{bmatrix}\right\}$
d) $\left\{\begin{bmatrix}2\\0\end{bmatrix}\right\}$
e) None of the above

1.5: If $A=\begin{bmatrix}-1 & 0\\ 0 & 1\end{bmatrix}$, then $A^9=$

a) $-A$
b) I_2
c) $-I_2$
d) A
e) None of the above

Q–2: [8 marks] Answer the following questions as True or False (2 mark), Justifying your answer (2 marks):

2.1: The linear system that has an augmented matrix $\left[\begin{array}{ccc|c}a & 4a & 4 & 8\\ 1 & 4 & 2 & 4\end{array}\right]$ has a unique solution at $a=-2$.

2.2: If $|A|=|B|$ then $|A-B|=0$.

Each question is worth 9 marks. Answer the following questions:

Q–3: [5+4 marks]

Let $W=\left\{\begin{bmatrix}a+3b\\ a-b\\ c\end{bmatrix}; c=5a+2b\ a,b\in R\right\}$ be a subset of $\boldsymbol{R^3}$.

a) Show that W is a subspace of $\boldsymbol{R^3}$.
b) Find the basis and the dimension of W.

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Q–4: [3+4+2 marks]

Define the mapping $T: R^3 \to R^3$ by $T\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x - z \\ y \\ x + z \end{bmatrix}$.

a) Show that T is a linear transformation.
b) Find the null of $T, N(T)$ and range of $T, R(T)$.
c) Find $Dim(N(T))$ and $Dim(R(T))$.

Q–5: [9 marks]

Let $A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$.

a) Show that $\lambda_1 = -1$, and $\lambda_2 = 3$ are eigenvalues of A.
b) Find matrices P and D such that $P^{-1}AP = D$; D is a diagonal matrix.

End of questions

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